

Security incident report

Section 1: Identify the network protocol involved in the incident

The incident involved DNS, TCP, and HTTP protocols. DNS was used to resolve `yummyrecipesforme.com` and `greatrecipesforme.com` into IP addresses. TCP established the connection between the browser and each web server using the three-way handshake, and HTTP was used to request and load the webpages. The traffic showed that the user was successfully connected to the legitimate website before being redirected to the malicious website.

Section 2: Document the incident

A former employee gained access to the website's administrative account by repeatedly trying known default passwords until the correct password was guessed. After accessing the admin panel, the attacker modified the website source code and added malicious JavaScript that prompted visitors to download an executable file. When the file was executed, the user's browser was redirected from `yummyrecipesforme.com` to the malicious site `greatrecipesforme.com`. Network traffic confirmed successful DNS lookups, TCP connections, and HTTP requests to both websites. The root cause was the use of a default administrative password and the absence of controls to prevent repeated login attempts.

Section 3: Recommend one remediation for brute force attacks

Implement a strong password policy that requires changing default credentials and using complex, unique passwords for administrative accounts. This should be combined with login rate limiting or account lockout controls to reduce the risk of brute-force attacks.

